

How do cars update firmware without bricking an ECU?

A complete software simulation of automotive OTA safety: signed packages, A/B flash slots, and automatic rollback.

Sakshi Anwekar × Tejas

github.com/sakshianwekar/sdv-ota-platform

Remote updates can fail unsafely

- Unsigned or tampered firmware can reach the vehicle
- Overwriting the running image risks a brick
- No automatic recovery when new firmware misbehaves

Real automotive software delivery

SDV platforms

Vehicles need continuous software improvement after sale.

ECU firmware OTA

Motor, brake, battery and other controllers update remotely.

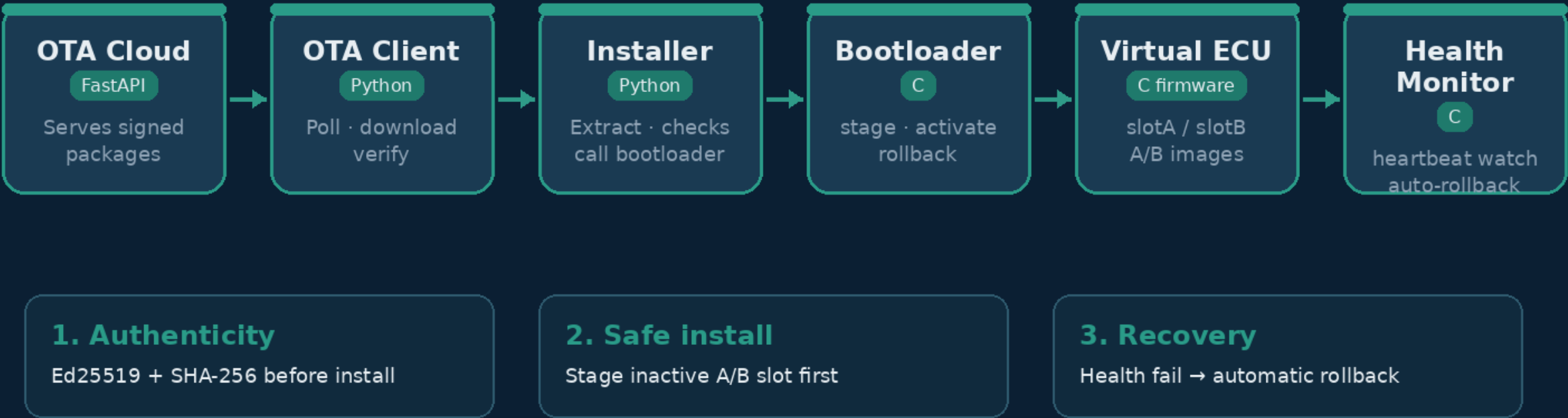
Fleet operations

Multiple ECUs / vehicles need the same safe update contract.

End-to-end OTA pipeline

SDV OTA Platform — Update Pipeline

Signed delivery · A/B flash slots · Health-checked auto-rollback



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Cloud → Client → Installer → Bootloader → ECU → Health Monitor

A/B flash slots — never overwrite running firmware

Active slot

Currently running
Stay online

Inactive slot

New package staged here
Activated only after verify

Bootloader stages into the inactive slot, then flips active on activate. Rollback flips back if health fails.

Cryptographic trust before flash

Ed25519 signature

Proves the package is authentic

SHA-256 checksum

Proves the bytes were not altered

Downgrade protection

Rejects older versions before staging

Tamper rejection

Invalid signature/checksum never reaches flash

End-to-end results

- **Healthy update**
v1.0 → v1.1 activates and stays healthy
- **Broken update**
Signed v1.2 fails health checks → auto-rollback
- **Security gates**
Tamper and downgrade rejected before flash
- **Fleet path**
Motor + Brake + Battery ECUs updated together

THE TEAM

Built by two engineers

Sakshi Anwekar

OTA Cloud (FastAPI), OTA Client,
Installer, packaging, security tests

Tejas

Bootloader (C), Health Monitor,
Virtual ECUs, firmware / fleet path

Want the repo and demo?

GitHub

github.com/sakshianwekar/sdv-ota-platform

Discussion question

In OTA systems, what matters more — cryptographic verification or runtime health checks? We implemented both as separate layers.

Open to conversations

Embedded · Backend · Automotive software

Links in the first comment on the LinkedIn post